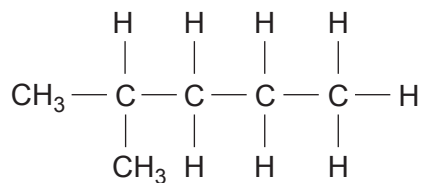


SECTION AAnswer **all** questions.

1. Give the name of the compound shown.

[1]

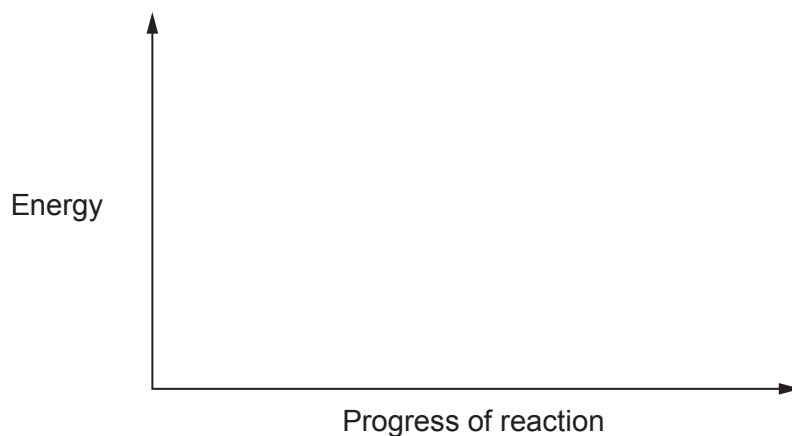


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2. Draw the energy profile diagram for an endothermic reaction.

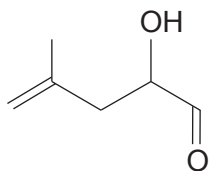
Label the activation energy, E_a , and the enthalpy change of the reaction, ΔH .

[2]



3. Give the molecular and empirical formulae of the compound shown.

[2]



Molecular formula

Empirical formula



4. Chloroethene is polymerised to make PVC.

(a) State why the chloroethene monomer does not exhibit *E–Z* isomerism. [1]

.....

.....

(b) Draw the repeating unit for PVC. [1]

5. Complete the equation to show the structure of the **main** product formed. [1]



6. Give the structure of a tertiary alcohol which contains 5 carbon atoms. [1]

7. Name the catalyst used in the hydrogenation of unsaturated oils to make saturated fats such as margarine. [1]

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